

Project Design Phase

Proposed Solution Template

Date	19 September 2026
Team ID	
Project Name	ShopEZ — Apparel E-Commerce Application
Maximum Marks	2 Marks

Proposed Solution Template

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Everyday clothing shoppers struggle to find accurate, size- and color-specific stock information and a trustworthy checkout, while store administrators lack a single, unified way to manage products, orders, and store content.
2.	Idea / Solution description	ShopEZ is a full-stack MERN application — a React (Vite) frontend, an Express/Node.js backend, and MongoDB Atlas for storage — that tracks inventory at the individual size and color variant level, and pairs a streamlined shopping experience with a complete admin dashboard for products, orders, and site settings.
3.	Novelty / Uniqueness	Unlike typical storefront builds that track a single stock number per product, ShopEZ models stock per variant (size + color) independently, and snapshots each order's item details (name, price, size, color) at the moment of purchase — so historical orders stay accurate even after a product is later edited, discounted, or removed.
4.	Social Impact / Customer Satisfaction	Shoppers gain confidence that what they order is genuinely available and can track its status from placement through delivery. Admins spend less time reconciling stock manually and can update categories or the homepage banner instantly, without a code change or deployment.
5.	Business Model (Revenue Model)	As a direct-to-consumer retail storefront, revenue comes from product sales. The same architecture could extend into commission-based multi-vendor listings, or a paid tier for the admin tooling itself if offered to other small store owners.
6.	Scalability of the Solution	The Express API is stateless (JWT-based authentication with no server-side session storage), so it can scale horizontally behind a load balancer. MongoDB Atlas' managed clusters scale independently of the application layer as the catalog and order volume grow.